

Geological Disposal Techniques for High-Level Nuclear Waste and their Impact on Environment: A Review

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The safe disposal of high-level nuclear waste (HLW), including spent nuclear fuel (SNF), is an inherent challenge for the sustainable use of nuclear energy. Deep geological repositories (DGRs) is the universally accepted solution for long-term isolation of HLW derived from biosphere. This review summarizes the current advances in DGRs, with an emphasis on site selection criteria, engineered barrier systems (EBS), and long-term safety assessments. Current trends in site characterization for host rock units that have low hydro-geological activity, whereas EBS designs include multi-layer containment materials such as bentonite clay, copper, and other geotechnic backfills in order to provide containment over geological timescales. Long-term safety assessments use innovative modeling methods to simulate radionuclide migration. Mitigation strategies tied to the established practices of engineered systems and natural analogues regarding biological and deep borehole disposal of nuclear waste are also discussed. This paper establishes the evolution of DGR practices, current challenges, and future directions in storage and disposal of HLW. Key considerations include site factors, high-level radioactive waste, engineering, site selection processes, engineered barrier systems, safety evaluations, disposal of boreholes.

Keywords: high-level nuclear waste, deep geological repositories, site selection, engineered barrier systems, safety assessments, borehole disposal

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1 Introduction

Nuclear waste is radioactive material left over from nuclear power generation, research, medical applications, or weapons production, requiring careful management due to its potential harm to humans and the environment. Its radioactivity, which stems from unstable atomic nuclei, can persist from days to millions of years, necessitating safe storage and disposal to prevent contamination.

Types of Nuclear Waste

1. **Low-Level Waste (LLW):** Includes items like contaminated clothing, tools, filters, and medical materials with low radioactivity.

Sources: Hospitals, research facilities, nuclear power plants, and decommissioning activities.

Management: Often compacted or incinerated, then disposed of in shallow land burial sites or stored until radioactivity decays to safe levels (days to years). Example: Gloves or rags used in nuclear facilities.

2. **Intermediate-Level Waste (ILW):** More radioactive than LLW, often requiring shielding but not cooling. Includes materials like reactor components, chemical sludges, or resins.

Sources: Nuclear power plant operations, fuel cycle processes, and decommissioning.

Management: Typically solidified (e.g., in concrete) and stored in engineered facilities or disposed of in intermediate-depth repositories. Example: Used reactor filters or contaminated metal parts.

3. **High-Level Waste (HLW):** Highly radioactive, heat-generating waste, primarily spent nuclear fuel or reprocessing byproducts.

Sources: Nuclear reactors and fuel repro-

cessing plants.

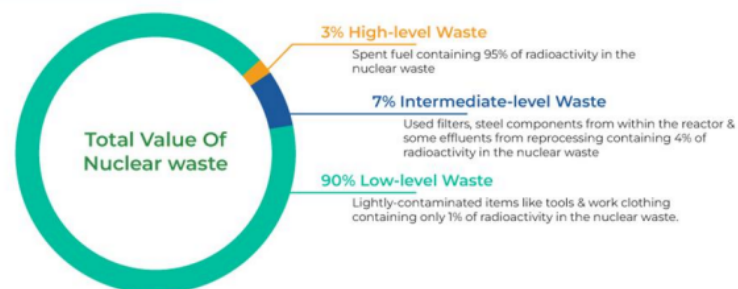
Management: Requires heavy shielding and cooling; stored in water pools or dry casks, with long-term plans for deep geological repositories. Can remain hazardous for thousands to millions of years. Example: Spent fuel rods from nuclear reactors.

4. **Transuranic Waste (TRU):** Waste containing elements heavier than uranium (e.g., plutonium), with long half-lives, typically from weapons production or research.

Sources: Nuclear weapons programs and some research reactors.

Management: Stored or disposed of in deep geological repositories due to long-term radioactivity. Example: Contaminated equipment from plutonium processing.

Types of Nuclear Waste



High-level nuclear waste (HLW), characterized by high activity concentrations and significant radiogenic heat, poses unique challenges for safe management due to its long-lived radionuclides, such as actinides (^{237}Np , ^{241}Am) with half-lives extending to millions of years. Disposal is viewed by the International Atomic Energy Agency (IAEA) as the emplacement of radioactive waste in a facility with no intent to retrieve, with deep geological repositories (DGRs) as the main alternative for HLW and SNF [2]. DGRs seek to enclose waste in depths of several hundreds of meters in stable geological formations, minimizing interaction with the biosphere. This summary

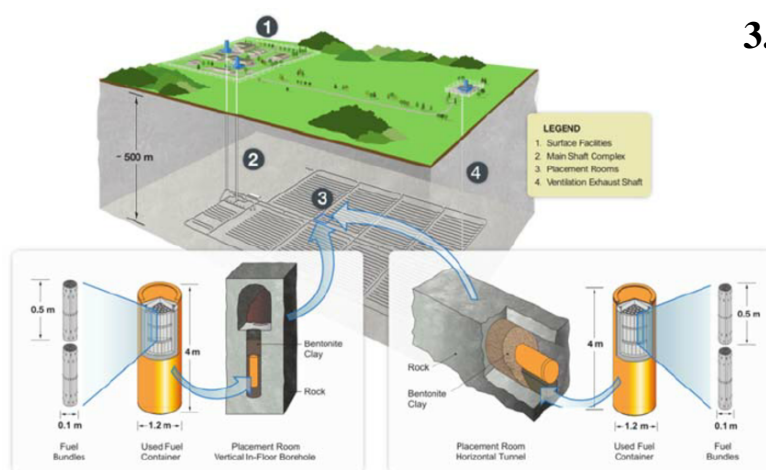
covers recent developments in DGR technologies, with on site selection criteria, engineered barrier systems (EBS), and long-term safety assessments simultaneously looking at emerging alternatives such as deep borehole disposal (DBD). HLW has a list of long-lived radionuclides such as plutonium-239, americium-241, and neptunium-237, with half-lives extending well into the thousands of years. The decay of these radionuclides' behavior poses severe threats to waste management because of the heat evolved. HLW generated by HLW requires proper storage and disposal planning. Proper HLW management is commonly done through deep geological burial in repositories specifically designed to separate the waste from the environment for thousands of years (United Nations, 2020).

2 History

In order to manage radioactive and hazardous waste, the idea of deep geological repositories first surfaced in the middle of the 20th century and underwent substantial development over time.

The concept's inception dates back to the 1950s. Deep geological disposal was first suggested in the 1950s. An important conceptual step was taken in 1957 when the US National Academy of Sciences assessed deep borehole disposal as a possible means of isolating nuclear waste. As nuclear energy expanded, the idea of isolating waste using stable geological formations gained popularity.

- **Early Implementation (1960s–1970s):** Experimental and operational facilities marked the beginning of practical steps toward DGRs. The Asse II salt mine in Germany was utilized as a research mine beginning in 1965. From 1967 to 1978, radioactive waste was deposited there, although leakage later became a problem.
- **Modern Development (1980s–Present):** In the 1980s and later, the DGR approach gained international traction, with several nations advancing construction and research: At Forsmark in crystalline rock, Sweden started running a DGR for low- and intermediate-level waste (L&ILW) in 1988. The most cutting-edge high-level waste (HLW) repository in Finland, Onkalo, was built in 2004 with KBS-3 technology and was originally scheduled to open in 2023 but was postponed. Although it manages defense-related transuranic waste rather than commercial spent nuclear fuel, the US's Waste Isolation Pilot Plant (WIPP), which has been in operation since 1999, is the only DGR presently in operation.
- **Formal Recognition and Policy (2000s–Present):** In numerous nations, the DGR approach emerged as the favored technique for managing nuclear waste: - DGR was the technical endpoint of the Adaptive Phased Management approach, which was suggested by Canada's Nuclear Waste Management Organization (NWMO) in 2004 and implemented by the government in 2007. International proposals for DGRs, like Australia's global repository, were debated in the early 2000s but encountered political opposition. By 2006-2007, DGR was formally recognized as a crucial radioactive waste management strategy by the European Union's CARD project and the Implementing Geological Disposal – Technology Platform (IGD-TP).



3 Environmental and Ecological impacts of radioactive waste disposal

3.1 Soil Contamination

1. Mechanisms of Radionuclide Deposition
Radionuclides enter soil through nuclear accidents, waste disposal, or atmospheric fall-out via wet (precipitation) or dry deposition. Soil properties like texture, pH, and organic matter influence radionuclide mobility. Sandy soils allow faster leaching, while clay soils retain radionuclides due to higher adsorption capacities.
2. Long-Term Effects on Soil Radionuclides
disrupt soil fertility by impairing microbial activity and nutrient cycling, reducing agricultural productivity. They create toxic environments, diminishing soil health and ecosystem stability.
3. Bioavailability in Agricultural Soil
Radionuclides like cesium-137 are more bioavailable in acidic soils, increasing crop uptake and food chain risks. Soil amendments, such as lime, can reduce bioavailability and mitigate contamination.

3.2 Case Studies

- **Chernobyl (1986):** High levels of cesium-137 and strontium-90 contaminated soils, altering microbial communities and reducing fertility.
- **Hanford Site (USA):** Plutonium and strontium contamination raised concerns about bioavailability and uptake by vegetation, impacting wildlife and humans.

3.3 Water Contamination and Groundwater Impact

4. Leaching into Water Sources
Radionuclides leach from contaminated soils into water through precipitation or irrigation. Cesium-137 and strontium-90 are highly mobile, influenced by soil composition and chemical properties.
5. Contamination of Surface Water
Runoff from contaminated areas affects rivers, lakes, and oceans, leading to bioaccumulation in aquatic organisms. The Chernobyl and Fukushima disasters caused significant contamination, impacting food webs and fisheries.
6. Groundwater Impact
Contaminated aquifers, critical for drinking water, remain degraded for decades or centuries due to radionuclides like uranium and tritium, posing health risks.

3.4 Airborne Radioactive Particles

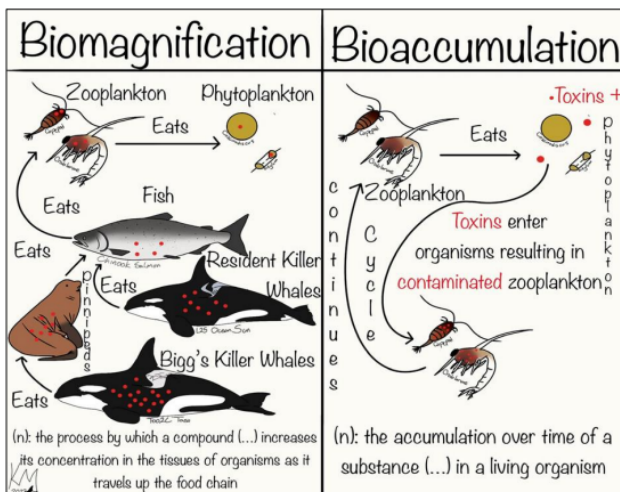
7. Dispersion Through Wind
Radioactive dust and particles, released during nuclear accidents or facility operations, spread via wind, contaminating distant areas. Isotopes like

cesium-137 and iodine-131 remain airborne for extended periods.

8. Atmospheric Effects Radioactive aerosols may alter local climates by affecting solar radiation absorption. Global dissemination, as seen in Chernobyl and Fukushima, raises international safety concerns.

4 Biological Effects of Radioactive Contamination

9. Bioaccumulation and Biomagnification
Bioaccumulation involves radionuclide buildup in organisms, while biomagnification increases concentrations up the food chain. Cesium-137 in Chernobyl's aquatic systems showed high concentrations in fish (e.g., carp, pike), posing health risks to humans.



10. Genetic Mutations in Flora and Fauna

- **Plants:** Radiation induces DNA mutations, altering morphology (e.g., leaf size) and reducing seed viability, as seen in Chernobyl's *Brassica rapa*.
- **Animals:** Radiation causes reproductive issues and malformations, e.g.,

limb deformities in Chernobyl's wild boars and reduced fertility in Japanese medaka fish. Transgenerational effects persist.

- **Examples:** Bluegill sunfish, northern leopard frogs, and house mice show genetic mutations affecting health and reproduction.

11. Impact on Biodiversity

- **Species Diversity Loss:** High radiation reduces plant and animal diversity, as seen in Chernobyl and Fukushima, due to mutations and reproductive failures.
- **Ecosystem Disruption:** Loss of key species alters food webs and nutrient cycling, reducing ecosystem resilience.
- **Reproductive Challenges:** Radiation causes reduced fertility and egg disruption in species like frogs and birds, increasing extinction.

5 Site Selection Criteria for DGRs

The selection of suitable sites for DGRs is critical to ensure long-term isolation of HLW. Key criteria, as outlined by the IAEA (3), include:

- **Geological Stability:** Host rock formations must have long-term stability against earthquake activity, erosion, and landslides. Common host rocks include crystalline rocks like granite, salt formations (e.g., Germany's Konrad repository), and consolidated clays (e.g., France's Bure site).
- **Hydrogeology:** Low groundwater flow is

essential to minimize radionuclide transport. Formations with stagnant pore water, such as clays, or no groundwater, like salt domes, are preferred .

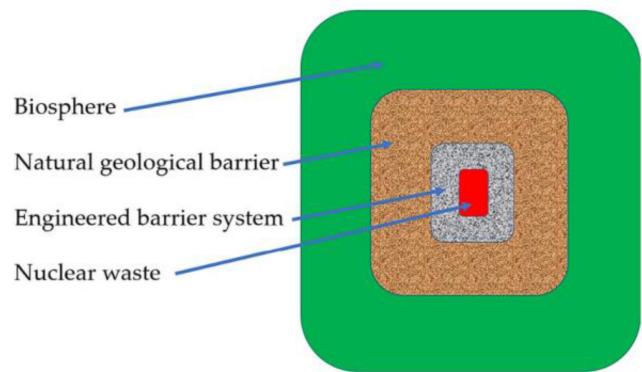
- **Geochemistry:** Reducing conditions that limit radionuclide solubility and migration are favored. For example, bentonite clays used in EBS can sorb radionuclides, enhancing containment.
- **Low Seismicity:** Sites with minimal earthquake risk are prioritized to prevent structural damage to the repository. Major underground test facilities are listed below.
- **Social and Regulatory Factors:** Public acceptance, land availability, and compliance with national regulations influence site selection. For instance, Sweden's Forsmark site was chosen after extensive stakeholder engagement.

Recent advancements in geology study include the use of advanced geophysical techniques, such as 3D seismic imaging and borehole logging, to characterize subsurface conditions with high precision. For example, Finland's Olkiluoto repository employs detailed seismic surveys to map fracture zones in granitic bedrock, ensuring very less groundwater interaction . Machine learning algorithms are also being explored to optimize site selection by predicting long-term geological stability based on historical data (11). **Deep geologic disposal has been studied for several decades, including laboratory tests, exploratory boreholes. Major underground test facilities are listed below.**

Country *	Facility name	Location	Geology	Depth *	Status
Belgium	HADES Underground Research Facility	Mol	plastic clay	223 m	in operation 1982 ^[12]
Canada	AECL Underground Research Laboratory	Pinawa	granite	420 m	1990–2006 ^[12]
Finland	Onkalo	Olkiluoto	granite	400 m	under construction ^[3]
France	Meuse/Haute Marne Underground Research Laboratory	Bure	claystone	500 m	in operation 1999 ^[13]
Japan	Horonobe Underground Research Lab	Horonobe	sedimentary rock	500 m	under construction ^[14]
Japan	Mizunami Underground Research Lab	Mizunami	granite	1000 m	under construction ^{[14][15]}
South Korea	KAERI Underground Research Tunnel	Deajeon	granite	120 m	in operation 2006 ^[16]
Sweden	Äspö Hard Rock Laboratory	Oskarshamn	granite	450 m	in operation 1995 ^[12]
Switzerland	Grimmel Test Site	Grimmel Pass	granite	450 m	in operation 1984 ^[12]
Switzerland	Mont Terri Rock Laboratory	Mont Terri	claystone	300 m	in operation 1996 ^[17]
United States	Yucca Mountain nuclear waste repository	Nevada	tuff, ignimbrite	50 m	1997–2008 ^[12]

6 Engineered Barrier Systems (EBS)

The EBS is a critical component of DGRs, designed to contain radionuclides until they decay to safe levels. The EBS typically includes the wasteform, container, backfill, and liners, working in concert with the natural barrier system (NBS) provided by the host rock (4). Recent developments in EBS components include:



- **Wasteforms** Durable wasteforms, such as borosilicate glass for HLW and cement for ILW, are designed to resist leaching over thousands to millions of years. Borosilicate glass, used in France and the UK, corrodes at rates of approximately $0.1 \mu\text{m/y}$ in non-saturated conditions, ensuring containment for up to a million years (6). Advances in glass formulations, such as iron phosphate glasses, show promise for improved chemical durability (7).

- **Containers** Containers provide physical containment and facilitate handling. Copper canisters with cast-iron inserts, as used in Sweden's KBS-3 concept, are designed to remain intact for 100,000 years due to their corrosion resistance in reducing environments (5). Recent innovations include the development of composite containers combining stainless steel and nickel-based alloys for enhanced durability in saline groundwater conditions (8).
- **Backfills and Buffers** Bentonite clay, widely used as a backfill material, swells upon contact with water, sealing voids and limiting radionuclide migration. Finland's Olkiluoto repository employs highly compacted bentonite to ensure long-term stability (1). Research into alternative backfill materials, such as geopolymers, aims to improve mechanical strength and radionuclide sorption capacity (7).
- **Lifetime Estimates** The lifetime of EBS components is estimated using corrosion rates and critical dimensions. For example, a 2 mm-thick carbon steel canister has a lifetime of approximately 660 years at a corrosion rate of 3 $\mu\text{m/y}$, while stainless steel canisters last significantly longer (9). These estimates inform the design of DGRs to ensure containment during the peak radiotoxicity period of HLW.
- **Natural Rock** The surrounding geology itself acts as a barrier, chosen for its stability and ability to trap any radioactive particles that might escape

DGRs and EBS are all about thinking long-term—like, really long-term. We're talking about keeping waste safe for longer than human civilization has existed. It's a huge responsibility, and

it's why every detail, from the type of rock to the thickness of a canister, is obsessed over. The stakes are high: a leak could contaminate groundwater or soil, though the multiple barriers make that extremely unlikely.

There's also the ethical side. Nuclear power helps us generate energy with low carbon emissions, but it leaves behind waste that future generations will inherit. DGRs and EBS are our way of saying, "We've got this covered," ensuring the waste stays out of harm's way, even when we're long gone.

7 Drawbacks of Current and Planned DGRs

1. **Geological Instability Risks Problem:** Geological structures need to be stable for thousands of millions of years of sands; yet natural events such as seismic, volcanic, or future glaciations can cause disruption of containment. For instance, WIPP's salt formation is stable but are prone to seismic activity, whereas Onkalo and Forsmark need to consider Glacial cycles influence hydrostatic pressure and groundwater movement. Effect: Potential for radionuclide release when barriers fail from unexpected geological alterations.
2. **Issue of Public and Political Opposition:** Public distrust, frequently driven by anti-nuclear sentiments, campaigns, and political opposition may delay or stop projects. Yucca Mountain was confronted strong opposition from Nevada's government and citizens, leading to its cancellation. In Germany, the Gorleben plant was protested against since it was near populated areas after reunification. Effect: Delays in site selection, licensing, and construction, which escalate expenses and extending surface

storage.

3. **Technical Problems: Corrosion and Barrier Failure**—Microbially induced corrosion. (MIC) through bacteria that produce corrosive metabolites (e.g., sulfides, organic acids) can Metal containers rust, especially in anaerobic conditions. For instance, copper can Storage in Forsmark and Onkalo is based on zero groundwater flow; however, microbial processes are able to compromise long-term integrity.
4. **Permanence vs. Retrievability:** Some designs allow retrieval of the waste for 100 years, which complicated and cost more to seal. Permanent sealing, as planned for WIPP, presents future unavailability if needed. Effect: Leaks or the requirement for expensive redesigns if screens perform poorly.
5. **Costly Delays and Excess Expenses Problem:** DGRs involve huge investment (billions of financial investment and long years of preparation, research, and development. For example, Yucca Mountain's research cost \$9 billion prior to cancellation, and Onkalo's construction costs decades. Effects: Economic cost to the government and utilities, with threats of financing abandonment or project cuts.
6. **Issues Involving Human Intrusion Threats:** Future human activities, e.g., drilling or mining, can unintentionally breach repositories. Germany's Asse II mine, used for LLW/ILW from 1967–1978, leaked radioactive brine through improper sealing, emphasizing intrusion hazards. Consequence: Possible environmental pollution in case the repositories are not clearly marked or safeguarded.
7. **Problem of Scalability and Limited Capacity:** DGRs such as Onkalo (6,500 metric tons) and Forsmark (12,000 metric tons) are small in size, insufficient to contain nuclear waste globally volumes. WIPP is limited to TRU waste, not HLW. Effect: Requirement for multi-repositories, rising costs and location issues.
8. **Social and Environmental Issue:** Construction disturbs natural habitats, and communities near planned sites will be concerned about environmental cognitive consequences or social shame. Public disillusionment, such as in Yucca Mountain, can derail projects. Impact: Resistance from indigenous groups, local residents, or environmentalists, delaying or halting projects.

8 Long-Term Safety Assessments

Long-term safety assessments, conducted through safety analysis reports (SARs), evaluate the radiological impacts of DGRs over geological timescales. The IAEA and International Commission on Radiological Protection (ICRP) set dose limits of 0.3 mSv/y for natural processes and 10 mSv/y for human intrusion, significantly below background radiation levels (1000–10,000 μ Sv/y) (10; 1).

8.1 Modeling Radionuclide Migration

Advanced modeling techniques, such as coupled hydrogeochemical models, simulate radionuclide transport through the EBS and NBS. For example, Monte Carlo simulations are used to account for uncertainties in groundwater flow and corrosion rates (11). Recent developments include the integration of machine learning to refine predic-

tive models, improving accuracy over million-year timescales.

8.2 Radiotoxicity and Retention Times

The relative radiotoxicity index (RTI) compares HLW toxicity to natural uranium ore, indicating that HLW reaches equivalent toxicity in approximately 3000 years, while SNF requires 100,000 years (5). Retention times are calculated using the equation:

$$\sum_{i=1}^N \frac{A_i \exp(-\lambda_i t_{\text{ret}})}{CL_i} = 1,$$

where A_i is the initial activity concentration, λ_i is the decay constant, and CL_i is the clearance level. For SNF containing ^{237}Np ($T_{1/2} = 2.1 \times 10^6$ years), retention times can exceed 42 million years (1).

8.3 Challenges and Innovations

Uncertainties in long-term predictions, such as climate cycles and geological changes, pose challenges for SARs. Emerging approaches include the use of natural analogues (e.g., uranium ore deposits) to validate models over geological timescales (13). Additionally, Generation IV reactors, which transmute minor actinides, could reduce HLW radiotoxicity, shortening required retention times (14). The disposal of nuclear waste poses considerable difficulties because of the waste's persistent radioactivity and possible environmental risks. Containment strategies that guarantee safety over geological timescales are necessary for high-level radioactive waste, such as spent nuclear fuel, which poses a threat for thousands to millions of years. Creating storage techniques that avoid soil and groundwater contamination or leakage is one of the main challenges because any failure could have detrimental effects

on the environment and human health. Long-term storage safety is complicated by geological uncertainties like earthquakes and groundwater movement. Furthermore, political obstacles prevent the construction of permanent repositories, and public opposition to waste disposal sites—fueled by mistrust and fear—often delays progress. Tight controls are also necessary due to security concerns about possible theft or misuse of radioactive material.” Deep geological repositories, which isolate waste in stable subterranean formations with engineered barriers, are one innovation that has emerged to address these issues. By changing long-lived isotopes into shorter-lived ones, advances in partitioning and transmutation technologies seek to decrease the longevity and toxicity of waste. Modular dry storage systems offer flexible and secure temporary solutions, while new materials like synthetic rock (Synroc) offer better containment. Safety is further improved by cutting-edge ideas like ultra-deep borehole disposal and the use of robotics for waste management and monitoring. There is also hope for secure and affordable disposal through international cooperation on shared repositories. Together, these developments seek to address the intricate, multidimensional issues surrounding nuclear waste management and guarantee public and environmental safety for coming generations.

9 Emerging Technologies: Deep Borehole Disposal (DBD)

Deep borehole disposal (DBD) has gained attention as a cost-effective alternative to DGRs for small volumes of HLW and disused sealed radioactive sources (DSRS). DBD involves drilling boreholes to depths of 1–5 km, placing waste in sealed containers, and backfilling with bentonite or cement (12). Advantages include reduced con-

struction times, lower costs, and minimal human intrusion risk. Countries like the UK, U.S., and Australia are exploring DBD, with pilot projects demonstrating feasibility in sedimentary and crystalline formations (11). However, challenges include ensuring borehole sealing and characterizing deep geological conditions.

10 Discussion

The advancements in DGRs reflect a mature approach to HLW disposal, with operational facilities like the Waste Isolation Pilot Plant (WIPP) in the U.S. and planned repositories in Sweden and Finland demonstrating technical feasibility (1). Site selection has become more precise with geophysical and computational tools, while EBS designs benefit from durable materials and innovative formulations. Long-term safety assessments are increasingly robust, though uncertainties over geological timescales remain a challenge. DBD offers a promising alternative for specific waste streams, but its scalability requires further validation.

11 Conclusion

Given the long-term dangers and environmental risks associated with radioactive materials, disposing of nuclear waste is a difficult task. The foundation for disposing of high-level waste (HLW) is still deep geological repositories, which depend on meticulous safety evaluations, sophisticated engineered barrier systems (EBS), and careful site selection to guarantee containment over thousands of years. The viability and security of nuclear waste management are being enhanced by advancements in materials science, predictive modeling, and alternative techniques like deep borehole disposal (DBD). In addition to technological advancements, successful implementation requires addressing political obstacles, security threats, and public concerns. Optimizing transmutation technologies to decrease waste longevity, scaling up DBD, and improving long-term predictive models to better manage uncertainties should be the top priorities of future research.

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